

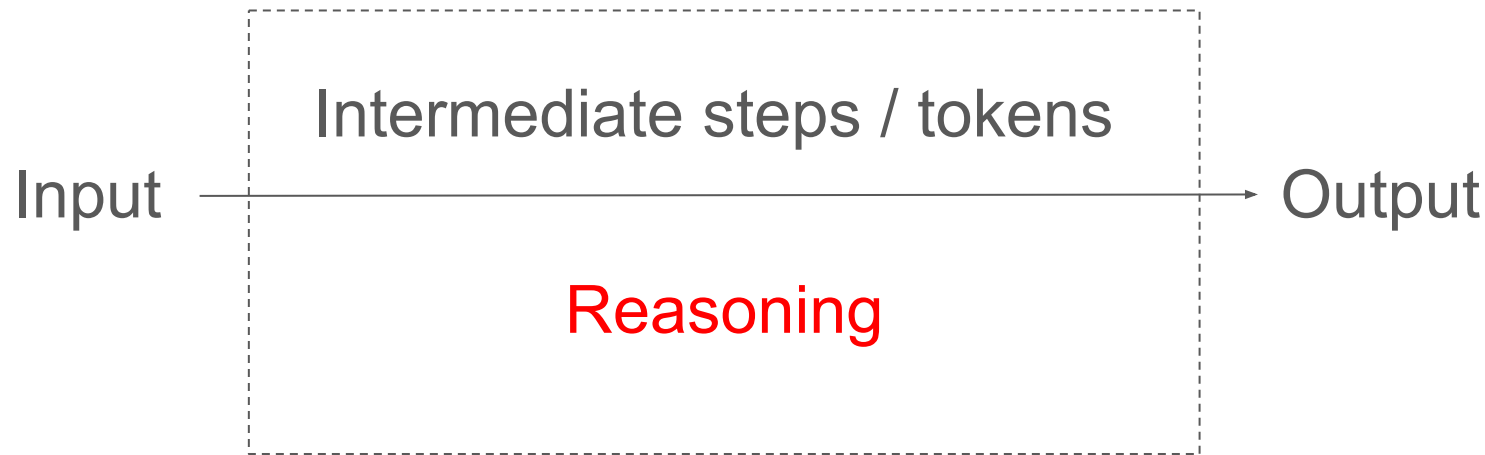
LLM Reasoning

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Google DeepMind

Stanford, CS25: Transformers United V5, April 29, 2025

What is LLM Reasoning?



The first ever paper which uses natural language to describe intermediate reasoning steps: Ling et al. Program Induction by Rationale Generation: Learning to Solve and Explain Algebraic Word Problems. ACL 2017. Before this paper, intermediate steps were described by symbolic languages (see the neural-symbolic reasoning literature, including my early work).

What is the output when concatenating the last letter of each word in “artificial intelligence”?



No reasoning

The answer is “le”.

Reasoning

The last letter of “artificial” is “l”. The last letter of “intelligence” is “e”. Concatenating “l” and “e” leads to “le”. So the answer is “le”.

This synthetic last-letter-concatenation task was motivated from my early work: Neuro-symbolic program synthesis. arXiv preprint 1611.01855 (2016). Joint work with Parisotto et al. At that time, we solved similar problems by automatic program synthesis.

Problem 1:

Question: Two trains running in opposite directions cross a man standing on the platform in 27 seconds and 17 seconds respectively and they cross each other in 23 seconds. The ratio of their speeds is:

Options: A) $3/7$ B) $3/2$ C) $3/88$ D) $3/8$ E) $2/2$

Rationale: Let the speeds of the two trains be x m/sec and y m/sec respectively. Then, length of the first train = $27x$ meters, and length of the second train = $17y$ meters. $(27x + 17y) / (x + y) = 23 \rightarrow 27x + 17y = 23x + 23y \rightarrow 4x = 6y \rightarrow x/y = 3/2$.

Correct Option: B

Why “Intermediate Tokens” / “Reasoning” Matters?

- For any problems solvable by boolean circuits of size T , **constant-size transformers** can solve it by generating $O(T)$ intermediate tokens
- If directly generating final answers, either requires a huge depth or cannot solve at all

Joint work Li et al. [Chain of Thought Empowers Transformers to Solve Inherently Serial Problems](#). ICLR 2024. “Chain of thought (CoT)” in this work just means “CoT reasoning”. Don’t mess it up with our “CoT prompting”.

Common Belief

Pretrained LLMs cannot reason without further prompting engineering or finetuning

WRONG

Pretrained LLMs are ready to reason

All we need is decoding

No reasoning? Check more generation candidates!

I have 3 apples. My dad has 2 more apples than me. How many apples do we have in total?

5 apples. (Greedy Decoding)

I have 3 apples, my dad has 2 more apples than me, so he has 5 apples. $3+5=8$.

We have 8 apples in total.

You have 3 apples, your dad has 2 more apples than you, so he has 5 apples. $3+5=8$.

The answer is 5.

How to select the best response? By length?

I have 3 apples. My dad has 2 more apples than me. How many apples do we have in total?

5 apples. (Greedy Decoding)

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The answer is 5.

Select responses with high confidence on answers!

I have 3 apples. My dad has 2 more apples than me. How many apples do we have in total?

5 apples. (Greedy Decoding)

I have 3 apples, my dad has 2 more apples than me, so he has 5 apples. $3+5=$ 8.

We have 8 apples in total.

You have 3 apples, your dad has 2 more apples than you, so he has 5 apples. $3+5=$ 8.

The answer is 5.

Way higher confidence on reasoning-based answers!

Chain-of-Thought Decoding

-
1. Go beyond greedy decoding by checking more generation candidates
 2. Choose candidates which have the highest confidence on the final answer
-

Can we **reshape the model's output distribution** so that thoughtful responses naturally rank 1st?

Chain-of-Thought Prompting

Q: Elsa has 3 apples. Anna has 2 more apples than Elsa. How many apples do they have together?

A: Anna has 2 more apples than Elsa. So Anna has $2 + 3 = 5$ apples. So Elsa and Anna have $3 + 5 = 8$ apples together.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Joint work with Wei et al. [Chain-of-thought prompting elicits reasoning in large language models](#). NeurIPS 2022

Let's Think Step by Step

The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Let's think step by step.

Kojima et al. Large language models are zero-shot reasoners. NeurIPS 2022.

Pros and Cons of Prompting

Pros: Simple and works

Cons:

CoT prompting needs task-specific examples

“Let’s think step by step” is generic, but performs much worse than few-shot

Pros and Cons of Prompting

Prompting approaches are actually weird

When asking someone a question —

will you first show similar problems/solutions before asking?

or, at the end of your question, will you have to say “let’s think step by step”?

Of course not!

Supervised Finetuning (SFT)

Step 1: collect a set of problems and their step-by-step solutions from human annotators

Step 2: maximize the likelihood of human solutions

Then apply the model everywhere

Ling et al. Program Induction by Rationale Generation: Learning to Solve and Explain Algebraic Word Problems. ACL 2017

Cobbe et al. Training Verifiers to Solve Math Word Problems. arXiv:2110.14168. 2021

Nye et al. Show Your Work: Scratchpads for Intermediate Computation with Language Models. arXiv:2112.00114, 2021

Supervised Finetuning (SFT)

What is the output when concatenating the last letter of each word in “artificial intelligence”? The last letter of “artificial” is “l”. The last letter of “intelligence” is “e”. Concatenating “l” and “e” leads to “le”. So the answer is “le”.

Elsa has 3 apples. Anna has 2 more apples than Elsa. How many apples do they have together? Anna has 2 more apples than Elsa. So Anna has $2 + 3 = 5$ apples. So Elsa and Anna have $3 + 5 = 8$ apples together.

Training data

Finetuning

LLM

Test problem

How many “r”s in “strawberry”?

Pros and Cons of SFT

Pros: Generic

Cons:

Does not generalize well

Scaling does not help much



How to Fix the Generalization Failure from SFT?

SFT procedure

Step 1: collect a set of problems and their step-by-step solutions from ~~human annotators~~

Step 2: maximize the likelihood of ~~human solutions~~

RL Finetuning

Repeat this process:

Step 1: collect a set of problems and their step-by-step solutions **generated from the model**

Step 2: maximize the likelihood of **correct solutions**

This technique were independently discovered by several labs. In Google DeepMind, the credit goes to Jonathan Lai and James An on my team. In the academia literature, the first paper of RL finetuning seems to be: Luong et al. ReFT: Reasoning with Reinforced Finetuning. arXiv:2401.08967. Jan 17, 2024.

RL Finetuning

Repeat this process:

Step 1: collect a set of problems and their step-by-step solutions **generated from the model**

Step 2: maximize the likelihood of **correct solutions**,
minimize the likelihood of wrong solutions

This technique were independently discovered by several labs. In Google DeepMind, the credit goes to Jonathan Lai and James An on my team. In the academia literature, the first paper of RL finetuning seems to be: Luong et al. ReFT: Reasoning with Reinforced Finetuning. arXiv:2401.08967. Jan 17, 2024.

Why “generated from the model” instead of “from humans”?

First principle in ML:

Directly optimize what we want!



So what do we want to optimize here?

Optimize “generation quality”!

Examples of quality measurement: correctness for solving math problems, BLUE score for machine translation, ...

All the rest is to compute gradients / backpropagation!

So what do we want to optimize here?

$$\max_{\theta} \mathbb{E}[r(\text{response}|\text{problem}, \theta)]$$

r measuring the quality of a response

θ model parameters

RL finetuning = Optimize it by gradient ascent

“Verification, the key to AI”

— Rich Sutton, 11/15/2001

A reliable verifier is the most crucial in RL finetuning, rather than the RL algorithm

Scaling Reasoning Training: What to Scale?

- For any problems solvable by boolean circuits of size T , **constant-size transformers** can solve it by generating $O(T)$ intermediate tokens — **Scaling output length!**
- If directly generating final answers, either requires a huge depth to solve or cannot solve at all — **Scaling model depth!**

Joint work with Li et al. [Chain of Thought Empowers Transformers to Solve Inherently Serial Problems](#). ICLR 2024. This theoretical work is for understanding the effectiveness of intermediate tokens (or chain of thought), and it has nothing to do with prompting (like our CoT prompting or let's think step by step).

The Beauty of LLM Reasoning

Human-like reasoning process emerges from token-to-token generation, rather than relying on exhaustive search as in classic AI

“Deep Blue was only intelligent the way your programmable alarm clock is intelligent”
— Garry Kasparov (after losing to Deep Blue in 1997)

[Question] Using the numbers from 1 to 10 to make 2025. Using each number once and primary operations + and *

[Answer] $(10 * 4 + 5) * (9 * 3 + 8 + 7 + 2 + 1)$

Let's break it down:

- First part: $10 * 4 + 5 = 40 + 5 = 45$
- Second part: $9 * 3 + 8 + 7 + 2 + 1 = 27 + 8 + 7 + 2 + 1 = 45$
- Final Calculation: $45 * 45 = 2025$

Therefore, the expression $(10 * 4 + 5) * (9 * 3 + 8 + 7 + 2 + 1)$ uses the numbers 1 through 10 exactly once and only the primary operations of addition (+) and multiplication (*) to equal 2025.

Gemini 2.0 thinking mode (Dec, 2024)

The target number is 2025. This is a relatively large number, suggesting multiplication will be heavily involved. It's also worth noting that 2025 is 45 squared ($45 * 45$). This might or might not be directly helpful, but it gives us a sense of scale. **Prioritize Multiplication:** Since the target is large, start thinking about how to get large intermediate products using multiplication. The largest numbers (10,9,8,7) are good candidates for multiplication early on ... This confirms multiplication is key, but simply multiplying everything together is too much. We need to combine additions. ... Let's aim for products that get us closer to the square root of 2025 (which is 45) ...

“We want AI agents that can discover like we can, not which contain what we have discovered. Building in our discoveries only makes it harder to see how the discovering process can be done.”

— Rich Sutton (“The Bitter Lesson”)

Pros and Cons of RL Finetuning

Pros: Generalize well for automatically verifiable tasks

Cons: Not all tasks are automatically verifiable

Further Great Improvement:
Aggregation and Retrieval

LLM reasoning is powerful, but any decoding issue in the paradigm of “generating reasoning tokens and then final answer”?



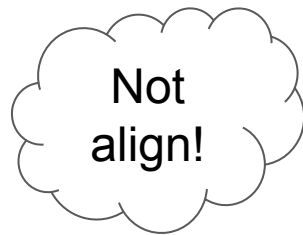
Always keep in mind that LLMs are probabilistic models trained to predict next tokens. **They are not humans!**

What LLM does in decoding:

$$\arg \max \mathbb{P}(\text{reasoning, final answer} | \text{problem})$$

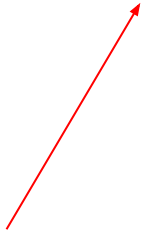
What we want:

$$\arg \max \mathbb{P}(\text{final answer} | \text{problem})$$



Let's fix it!

Marginalization



$$\begin{aligned} & \arg \max \mathbb{P}(\text{final answer} | \text{problem}) \\ &= \sum_{\text{reasoning}} \mathbb{P}(\text{reasoning, final answer} | \text{problem}) \\ &\approx \frac{\text{frequency of final answer}}{\text{total number of sampled responses}} \end{aligned}$$

Self-Consistency

1. Generate multiple responses by randomly sampling
 2. Choose the answer that appears most frequently
-

[Question] Janet's ducks lay 16 eggs per day. She eats three for breakfast every morning and bakes muffins for her friends every day with four. She sells the remainder for \$2 per egg. How much does she make every day?

Sampled responses:

Response 1: She has $16 - 3 - 4 = 9$ eggs left. So she makes $\$2 * 9 = \18 per day.

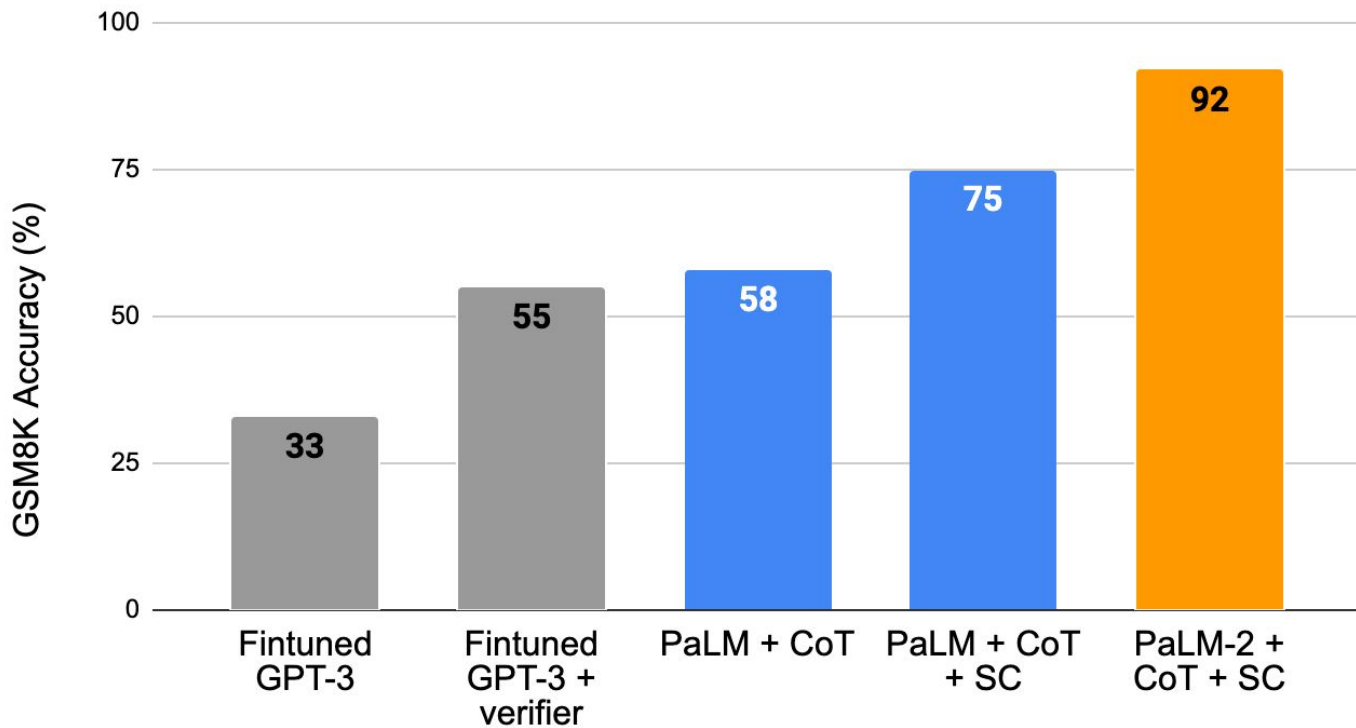
Response 2: This means she she sells the remainder for $\$2 * (16 - 4 - 3) = \26 per day.

Response 3: She eats 3 for breakfast, so she has $16 - 3 = 13$ left. Then she bakes muffins, so she has $13 - 4 = 9$ eggs left. So she has $9 \text{ eggs} * \$2 = \18 .

Most frequent answer is: 18
(Not most frequent reasoning path!)



Results on GSM8K (8 shots, Jan 2022/3)

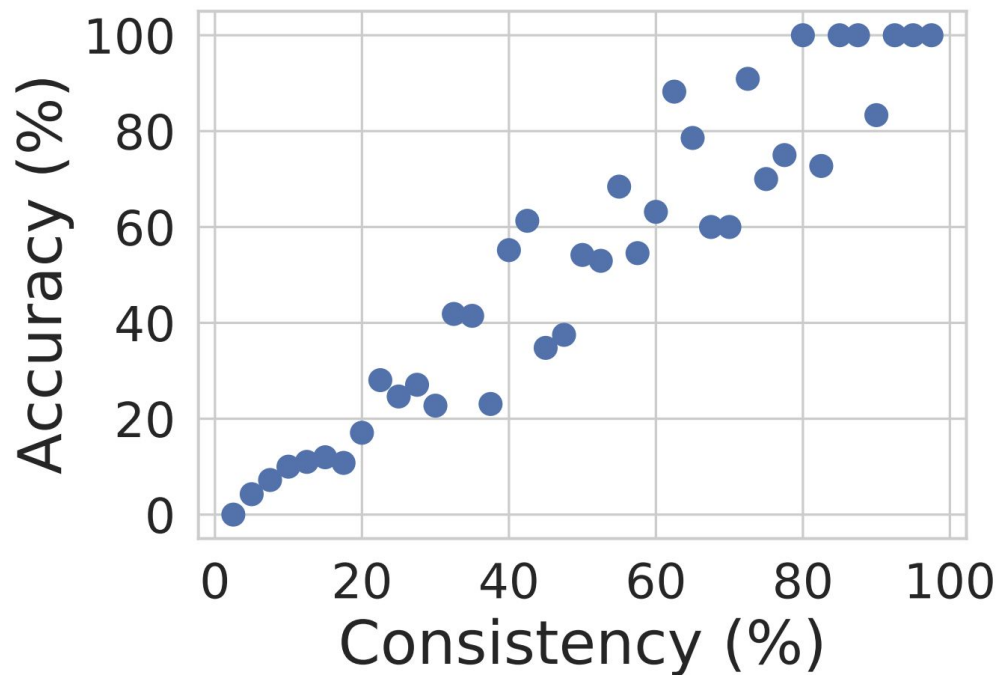


Appendix A

Dataset	Metric	gpt-4o	o1-preview	o1
Competition Math	cons@64	13.4	56.7	83.3
	pass@1	9.3	44.6	74.4

<https://openai.com/index/learning-to-reason-with-llms/>

Higher Consistency Indicates Higher Accuracy



QUIZ

[Q1] When the LLM outputs a direct answer without intermediate steps, will you still sample several times, and then choose the most common answer?

[Q2] Change self-consistency by letting LLM generate multiple responses, instead of sampling multiple times, and then choosing the most common answer. Does this make sense?



$$\arg \max \mathbb{P}(\text{final answer} | \text{problem})$$

How about free-from answers?

Universal Self-Consistency (USC)

Ask LLMs to self-select the most consistent answer

[Question] Where do people drink less coffee than they do in Mexico?

Response 1: ... Some examples include Japan, China and the United Kingdom.

It is important to note that coffee consumption can vary among individuals within these countries, and preferences can change depending on different factors such as...

Response 2: People in countries like Japan, China, and India typically drink less coffee than they do in Mexico...

Response 3: There are several countries where people generally drink less coffee compared to Mexico. Some of these countries include:

1. Japan:...
2. China...
3. Saudi Arabia...
4. India...

...

The most consistent response: 2

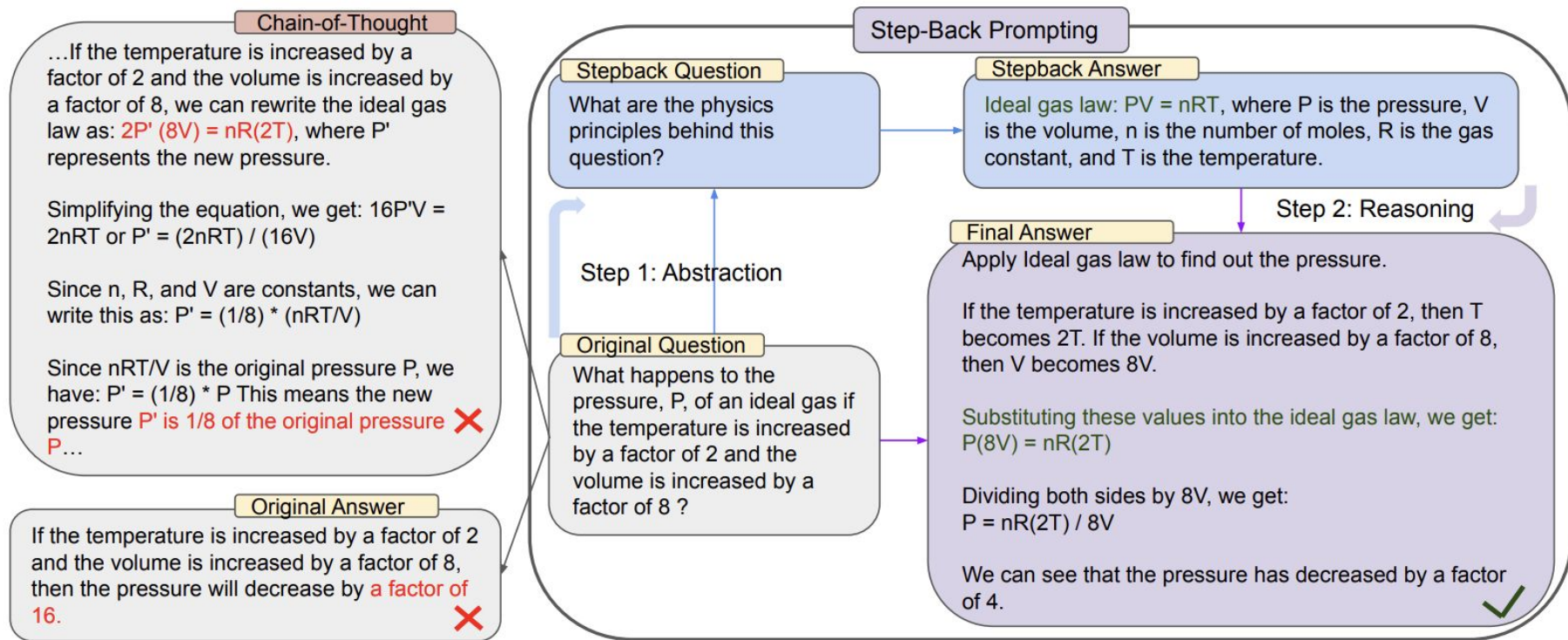
Retrieval or reasoning?

Do “retrieval + reasoning”!

What is the area of the square with the four vertices at $(-2, 2)$, $(2, -2)$, $(-2, -6)$, and $(-6, -2)$? Recall a related problem, and then solve this one.

The related problem is finding the distance between two points on a coordinate plane. To find the distance between two points, we use the distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

In this problem, we have four points: $(-2, 2)$, $(2, -2)$, $(-2, -6)$, and $(-6, -2)$. Let's choose $(-2, 2)$ and $(2, -2)$ to find the distance d between them: $d = \sqrt{((2 - (-2)))^2 + (-2 - 2)^2} = \sqrt{(4)^2 + (-4)^2} = \sqrt{(16 + 16)} = \sqrt{32}$. We know that the distance between two opposite sides of a square is equal. So, the length of each side of the square is $\sqrt{32}$. To find the area of the square, we square the length of one side: $\text{Area} = (\sqrt{32})^2 = 32$.



[Take a step back: Evoking reasoning via abstraction in large language models.](#) ICLR 2024. Joint work with Zheng et al.

Gemini Deep Research



Search



Deep research



Summary

- Reasoning > no reasoning
- RL finetuning > SFT
- Aggregating multiple answers > single answer
- Retrieval + reasoning > reasoning only

Next Big Breakthroughs

Solve the tasks beyond unique verifiable answers

Build real applications rather than solving benchmarks

THE END

“The truth always turns out to be simpler than you thought.”
— Richard P. Feynman